

Martin Beneš

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Experience

09 – 10 2025	Visiting Scholar , Binghamton University Grant application UIBK Travel grant
07 – 09 2024	Intern/R&D , project Stealthy pixels, Blindspot Solutions s.r.o. tatouage [Python] Python package implementing modern watermarking. Techtalks Series of presentations on steganography/steganalysis and watermarking.
Since 09 2021	Research Assistant , Security and Privacy Lab, University of Innsbruck UNCOVER [Horizon 2020 project] Developing a steganalysis framework for law-enforcement agencies. <i>Handling confidential data.</i> jpeglib [Python, C++] Python package enveloping C library libjpeg. conseal, sealwatch [Python, Rust] Python packages implementing modern steganography and steganalysis. Lecturer Information hiding (2025, 2026) · Watermarking and fingerprinting for natural and generated media. <i>Designing/conducting the courses.</i> Guest lecturer Real-world steganography and steganalysis. {University of Twente} Panel respondent Responsibility. {Austrian Computer Science Day} Workshop [Python] Drum synthesizer. {InDay} Workshop [Python] Digital Magic Ink. {Tag der MIP} · {GYREC} Workshop [C++] Steganography in a video stream. {InDay} · {LNDF}
05 2020, 09 2020	Core-team Contributor , COVID-19 Data Hub (JOSS paper, Nature paper), Emanuele Guidotti Data Hub [R, Python] Open data collection and cleaning.
09 2019 – 06 2021	Backend Developer , later QA Engineer/Scrum master , AB TaxiCaller Nordic OpenStreetMap solution [Python, MariaDB, Node.js, Pelias] maptiles server, geocoding, address interpolation. BloodHub integration [Node.js] Integration of blood delivery system. API [Jackson, OpenAPI, TypeScript] Rewriting and documenting. Revenu Quebec certification [Java/Android] Sales-recording system. Scrum master [Jira] Sprint planning. Driver app [React.js] Tester, iterative development. QA engineer [Node.js, Jenkins] Integration and deployment testing.
01 2017 – 01 2019	Software Developer , Artisys s.r.o. Air-traffic control display panel [C++/Qt5] Display panel for visualizing the radar targets.

Education

Since 10 2021	PhD., Computer Science , University of Innsbruck <GPA 3.64> Paper Variational cover modification steganography. {NeurIPS 2026} Paper Automating color image steganography research from first principles. {ESORICS 2026} Paper DCTdiff: Intriguing properties of image generative modeling in the DCT space. {PMLR ICML 2025} Paper Kerckhoffs in prison: A study of the steganalyst's knowledge. {IEEE WIFS 2024} Paper Exploring diffusion-inspired pixel predictors for WS steganalysis. {ACM IH&MMSec 2024} Paper Cover-source mismatch in steganalysis: Systematic review. {EURASIP JIS 2024} Paper JPEG steganalysis using leaked cover thumbnail. {IEEE WIFS 2023} Paper The effect of the JPEG implementation on the cover-source mismatch error in image steganalysis. {EUSIPCO 2022} Paper Know your library: How the libjpeg version affects compression and decompression results. {ACM IH&MMSec 2022} Reviewer Springer ESORICS · IEEE TIFS · IEEE WIFS · IEEE SPL · IEEE TDSC · AEEE Master thesis supervision Padding attacks against steganography in JPEG images. Master thesis supervision Demosaicing impact on forensics: How color channel dependencies compromise PRNU. Bachelor thesis supervision Accelerated implementation of syndrome-trellis codes for steganography. Summer school Steganography and steganalysis (lecturer, CTF team lead) · Privacy-preserving machine learning
06 2021	MSc., Statistics and Machine Learning , Linköping University <GPA 3.09> Master thesis [Python] HMM Modelling of the Spread of the SARS-CoV-2. {NordicMathCovid Oslo Workshop 2023} Research project [Python] Covid-19 regional comparison. Teacher assistant [R] Advanced R Programming, Computational Statistics, Machine Learning. <i>Conducting the seminars.</i>
08 2019	Bc., Information Technology , Brno University of Technology <GPA 3.11> Bachelor thesis [Arduino/C++, Python] Counting People Using PIR Sensor. <i>Joint research with FH Vorarlberg.</i> Erasmus Computer Science, FH Vorarlberg.
06 2016	Gymnázium Brno-Řečkovice . High school. Final <i>maturita</i> exam: Czech, Math, IT. <cum laude>

Skills

Computer security	Steganography and steganalysis · Digital watermarking · Multimedia forensics · Adversarial ML · Privacy-preserving ML
Data science	Statistical/probabilistic modeling (machine learning, deep learning, bayesian statistics) · Adversarial optimization · Computer vision Signal processing and series analysis (spectral analysis, source/channel coding, time-series forecasting) · Data mining and visualization MLOps (W&B) · Data engineering · SQL (MySQL, PostgreSQL, SQLite) · NoSQL (MongoDB, Redis) · Database modelling (UML)
Computing	HPC (Slurm) · Parallel programming (MPI) · Acceleration (GPU, multi-GPU, Ray, JIT) · Cloud computing (Docker, AWS, Terraform)
Software development	Python (PyTorch, TensorFlow, scikit-learn) · R · C/C++ · Bash · Networking (socket, openAPI) · Web (Node.js, jQuery, web scraping) IoT (Arduino, Raspberry Pi) · Hardware (FPGA/VHDL, x86 Assembly) · Project management (Scrum/Jira, Git, Jenkins)
Multimedia	TeX/Tikz · Typography · Graphics (CorelDraw/Inkscape, Photoshop) · Video production and editing (GoPro, DaVinci, Vegas) Sound processing (Cubase, Audacity) · Audio synthesis · Music theory
Languages	Czech/Slovak (native) · English (C1/CAE) · German (B2/ÖSD, Kurrentschrift) · Russian (~B2) · Polish (~B2) · Swedish (B1) · French (~B1)
Academia and research	Research (empirical, fundamental) · Experimental design · Critical assessment (peer review) · Academic and technical writing Scientific communication · Grant preparation · Teaching and mentoring · Course curriculum design · Thesis supervision
Genealogy	Record analysis (civil, church, military) · Historical reasoning · Qualitative interviews · Ancestral and chronological reconstruction
Other skills	Intercultural communication · Adaptivity · Creativity · Driving license B (car) · Ultralight backpacking · Bass guitar

Other projects and work

11 2026	Kterak se Cihláři dostali do Bosonoh Brochure about the family history.
10 2025 – 01 2026	Participant, AI Act's Code of Conduct, EU AI Office Working group on transparency for the GenAI providers.
08 2022	Jura jede [D3.js/jQuery] Data dashboard.
03 2021, 10 2021	Data engineer, University of Neuchâtel [Python] Database merging.
06 – 09 2020	Data engineer, Linköping University [Python] project M/F Ratio of current excess mortality, data collection and cleaning.
09 2018 – 12 2019	Tutor, Top Learning s.r.o. Individual tutoring in math and physics.
09 2013 – 03 2018	Sound engineer, ZiveOzvuceni.cz Installation and configuration of sound systems on various live events.
06 – 09 2018	Stock lead, I Love Sugar Work & Travel J1 program in USA. Managing stock and employees.
08 2016/2017/2018	Vice-manager, GenAgro a.s. Managing a large campsite for the Grand Prix motorbike race and its employees.